High Luminescent Red Quantum Dot Light-Emitting Diodes by Inkjet Printing

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Abstract

We report a high luminescent quantum dot light-emitting diode by inkjet printing. The relationship between film morphology versus the ink formulation and the annealing were investigated. For the best ink formulation among a few dozen devices we studied, the luminance is as high as 42 000 cd/m^2, and the current efficiency reaches 6.42 cd/A which is the highest reported for a very primitive device structure.

Author Keywords

QLED, inkjet printing, QD ink

1. Introduction

Colloidal quantum dots light-emitting diodes (QLED) have achieved great attention due to their outstanding property, like size controlled emission wavelengths, narrow emission bandwidths, and solution-processability.[1–4] Since the first report for QLED is in the 1990s, the EL and efficiency have a great progress during past 20 years.[5] Up to date, the champion device is comparable to the organic LEDs.[6–8] However, the QLED with high luminance and efficiency were mainly fabricated by spin coating.[9] The process is not amicable to mass production, moreover, spin coating results in loss of more than 94% of QDs solution.[10] Other film deposition techniques are still under developing to fulfill the demand for fabricating patterned devices with R, G, B colored QDs. Very recently, Li et al. reported an approach by a fibrous liquid bridge to form uniform QD film, which reduce the loss of QD solution.[11] However, similar as spin coating, it is still difficult to scale up, which limits the business application of QLED in the display field.

Inkjet printing is particularly attractive for its advantages like no-contact process, on-demand jetting, mask-free, and flexible substrate capability.[12–14] In 2009, Jabbour et al. first report the fabrication of QLED by inkjet printing.[15] However, the EL and efficiency are not very ideal, mainly because the poor film quality of QD layers.[16] Meanwhile, another problem is that the ink of QD is very different from the solvent for spin coating, because of the limitation of inkjet printer.[12,17] Besides the solubility of QD needs to be considered, in some extreme conditions, the printed inks could dissolve the hole transporting layer (HTL) for the normal non-inverted device structure. Xie et al. has tried to overcome this issue by adopting other corsslinkable hole transporting materials to against the solvent. [18] The discussion on the ink formula has not been proceeded since the work of Jabbour. [15] Therefore, it is still very crucial to develop a novel QD ink and corresponding processing methodology to obtain uniform QD layer for QLED.

As traditional HTL material, poly[(9,9'- dioctylfluorenyl- 2,7-diyl)- co-(4,4'-(N- (4- sec-butyl)) diphenylamine)] is commonly adopted in QLED to achieve long device lifetime, however the TFB layer damage in inkjet-printed device have to be settled urgently.[19–21] Herein, we report a good performance red QLED fabricated from such an ink inkjet-printed on TFB film. The ink is harmless for TFB compared to the reported inks. The red QLED exhibit a maximum luminance of 42 000 cd/ m^2 and a peak current efficiency of 6.4 cd/A which is the highest reported in the literature.

2. Result and discussion

As already reported in the literature,[22] cyclohexylbenzene (CHB) and decane are used here to prepare a printable QD ink as a control study. Since all the component here can dissolve QD well, the ink parameters could be easily adjusted by varying each component without disturbing the printability. For our novel ink, the a traditional QD solvent (QDS) serve as the main solvent and an additive is used to adjust the viscosity of the QDS. The inks were prepared with QD concentration 15 mg/ml, and the relationship between viscosity and surface tension as component variation are shown in Table 1. As reported that 9 : 1 for CHB and decane are suitable to apply on inkjet printing [22], the ink of 90 % CHB (noted as Ink 1) is chosen to study as a example with viscosity of 1.54 cP and 32.6 mN/m. For our ink formula, ink with 70% QDS presents the similar viscosity about 1.22 cP and much lower surface tension (23.7 mN/m), which is noted as Ink 2. Under applied voltages, as shown in Fig. 1(a), Ink 1 can form stable droplet under 30 V driving voltage (Fig. 1(b)). Due to the low surface tension of Ink 2, the driving voltage is slightly adjusted to 20 V to form stable droplet and eliminate satellite effect (Fig. 1(d)). The both inks were printed on TFB film and two evaporation types were chosen: heating (H) and vacuum (V). To study the effect for TFB, the blank solvents of Ink 1 and Ink 2 were used to wash TFB film, and the absorption spectrum of TFB layer after washing are shown in Fig. 2. Visibly, the absorption of TFB reduce significantly after ink 1 washing due to the existence of phenyl group, which indicates that Ink 1 can seriously damage TFB layer, but the absorption spectrum of TFB is not changed after Ink 2 washing.

Table. 1 Viscosity and surface tension of QD inks with component variation.

CHB %	90 (Ink 1)	70	60
Viscosity (cP)	1.54	1.11	1.02
Surface tension(mN/m)	32.6	29.7	27.9
QDS %	90	70 (Ink 2)	60
Viscosity (cP)	0.72	1.22	1.92
Surface tension (mN/m)	22.9	23.7	24.9

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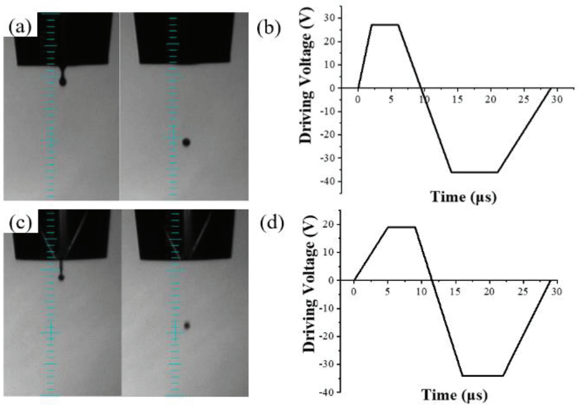

Fig. 1 (a) Jetting behaviors and (b) the driving voltage of Ink 1; (c) jetting behaviors and (d) the driving voltage of Ink 2 as time elapses.

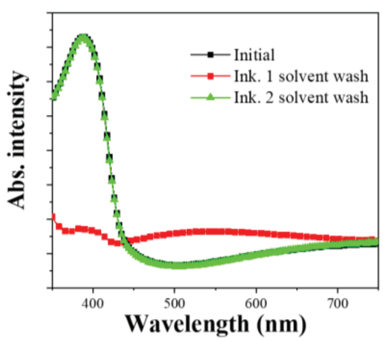

Fig. 2 UV-vis absorption spectra of TFB layers with Ink 1 and Ink 2 solvent rinsing.

The microscopic images of printed patterns on the TFB layer also verify the above assumption (Fig. 3). Apparently, not matter what processing method is applied on the printed film, Ink1 always present obvious coffee-ring effect and a dissolution of TFB layer. But the dots of Ink 2 demonstrate a more uniform surface on TFB because its lower surface tension and non interfering the bottom layer.[23,24] However, comparing to the heating method, the dots made by Ink 2 using vacuum evaporation show condensed and evenly spread surface, while the Ink 2 using heating method demonstrated a slighter coffee ring by due to the fast evaporation rate of solvent on a hotplate.[25]

For fabrication of QLED, both inks with pitch of 80 mm were inkjet printed in dot arrays on the surface of TFB as the emitting layer. The printed layers with heating or vacuum processing are marked as Ink 1-H, Ink 1-V, Ink 2-H and Ink 2-V. The atomic force microscopy (AFM) height images and the corresponding 3D images are shown in Figure 3. The RMS values for the four types of films are 24.49 nm, 9.87 nm, 1.72 nm, and 0.92 nm, respectively. The different RMS value suggests that the film qualities are significantly affected by the processing conditions for various inks. The Ink 1 films show rougher compared to Ink 2, because CHB can partially redissolve TFB layers, which leads to a high RMS value of TFB films and so does the above QDs layers, which may lead to pinholes and unevenly charge transfer. Accordingly, the Ink 2-V film shows the most flat and smooth surface than the others, which is a foreseeable good indication for inkjet printed QLED.

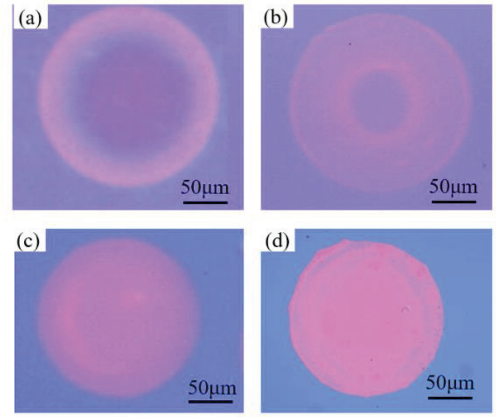

Fig. 3 The fluorescent microscopic images of printed dot using (a) Ink 1-H, (b) Ink 1-V, (c) Ink 2-H and (d) Ink 2-V.

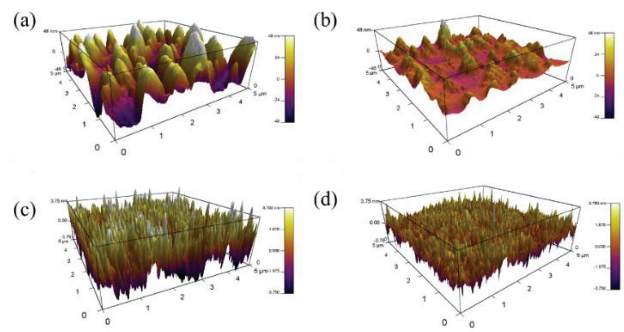

Fig. 4 AFM images of inkjet printed QDs layers using (a) Ink 1-H, (b) Ink 1-V, (a) Ink 2-H and (a) Ink 2-V.

The QLEDs are fabricated with a multilayers structure of ITO/ PEDOT:PSS (40 nm)/ TFB (30 nm)/ R-QD (inkjet printed)/ ZnMgO (noted ZMO;40 nm)/ Al (100 nm) (Fig. 5(a)), and the energy level of each functional layer is shown in Fig. 5(b). The electroluminescence (EL) spectra peak of all red film are located at ~ 630 nm (Fig 5(c)). Fig. 5(d-f) demonstrate the current-density-voltage (J -L-V) characteristics and the current efficiency of the four types of red inkjet printed QLEDs, and the detailed EL characteristics of devices are summarized in Table. 2. The Best inkjet printed device (Ink 2-V) exhibit maximum electrical luminance (EL) and maximum current efficiency (CE) of 42000 cd m^{-2} and 6.42 cd/A for Ink 2-V device, which is over 100 times higher than the control samples. The poor performance of Ink 1 is attributed to the dissolution of TFB layer leading to a large leakage current density, as we have shown in the above discussion. Without new ink recipe, there is almost no interfering between the QD layer and hole transporting layer, which could grantee the carrier transportation smoothly. For the same type of ink, with heating evaporation, the performance of QLED is 23431 cd/m^2 and 4.56 cd/A, Vacuum processing has significantly improved the QLED efficiency further by 40%. The significant QLED performance causes by the smooth QD layer and the non interfering the TFB layer.

After browsing all reported inkjet printed red QLEDs (EL peak: 600 ~ 650 nm) (Table 3), it is worth noting the Ink 2-V device is the best inkjet printed QLED reported in the literature. For the similar inkjet printed devices using TFB as hole transporting layer, this result is two order of magnitude higher than that of the same device structure.

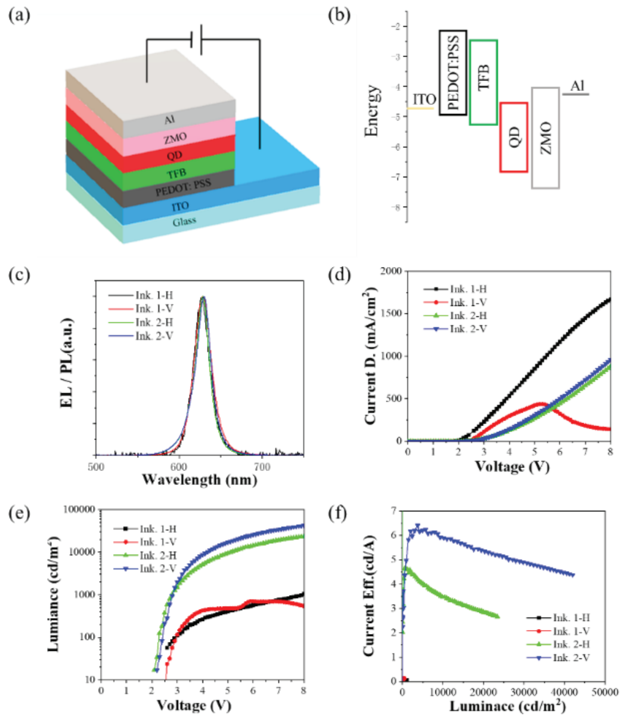

Fig. 5 (a) Schematic of the QLED device structure. (b) Schematic energy level diagram of QLEDs. (c) Normalized EL spectrum of QLED devices. Performance of the QLED devices: (d) current density-voltage (J–V) characteristics, (e) luminance-voltage (L–V) characteristics, and (f) current efficiency as a function of current density.

Table. 2 Summary of EL characteristics of the devices.

	Max CD (mA/m^2)	Max EL (cd/m^2)	Max CE (cd/A)
Ink 1-H	1667	1031	0.05
Ink 1-V	490	707	0.14
Ink 2-H	882	23431	4.56
Ink 2-V	958	42000	6.42

3. Conclusion

In summary, our work demonstrates the poor performance of inkjet printed QLED caused by the inks interfering the HTL and the rough QD layer, so we adjusted the ink recipe and evaporation condition to improve the QD/TFB interface and the QD layer. As a result, the QLED exhibit an average current efficiency of 6.42 cd/A and luminance of 42 000 cd/m^2, which is the highest value reported for the inkjet printed QLED. Our in-depth study show that inkjet-printing is an attractive digital printing technology for cost-effective, environmentally friendly integration of QLED. We believe that these printed light emitting devices could allow for full integration into the manufacturing process.

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References

[1] J. Shamsi, A. S. Urban, M. Imran, L. De Trizio, L. Manna, *Chem. Rev.* **2019**, *119*, 3296.

[2] H. Shen, W. Cao, N. T. Shewmon, C. Yang, L. S. Li, J. Xue, *Nano Lett.* **2015**, *15*, 1211.

[3] S. Kim, S. H. Im, S. W. Kim, *Nanoscale* **2013**, *5*, 5205.

[4] S. Pradhan, F. Di Stasio, Y. Bi, S. Gupta, S. Christodoulou, A. Stavrinadis, G. Konstantatos, *Nat. Nanotechnol.* **2019**, *14*, 72.

[5] V. L. Colvin, M. C. Schlamp, A. P. Alivisatos, *Nature* **1994**, *370*, 354.

[6] H. Zhang, X. Sun, S. Chen, *Adv. Funct. Mater.* **2017**, *27*, 1.

[7] X. Qu, N. Zhang, R. Cai, B. Kang, S. Chen, B. Xu, K. Wang, X. W. Sun, *Appl. Phys. Lett.* **2019**, *114*, 1.

[8] W. Cao, C. Xiang, Y. Yang, Q. Chen, L. Chen, X. Yan, L. Qian, *Nat. Commun.* **2018**, *9*, 2.

[9] Y. Shirasaki, G. J. Supran, M. G. Bawendi, V. Bulović, *Nat. Photonics* **2013**, *7*, 13.

[10] T. Zhu, K. Shanmugasundaram, S. C. Price, J. Ruzyllo, F. Zhang, J. Xu, S. E. Mohney, Q. Zhang, A. Y. Wang, *Appl. Phys. Lett.* **2008**, *92*, 10.

[11] X. Li, B. Hu, M. Zhang, X. Wang, L. Chen, A. Wang, Y. Wang, Z. Du, L. Jiang, H. Liu, *Adv. Mater.* **2019**, *1904610*, 1.

[12] M. Singh, H. M. Haverinen, P. Dhagat, G. E. Jabbour, *Adv. Mater.* **2010**, *22*, 673.

[13] Z. Zhan, J. An, Y. Wei, V. T. Tran, H. Du, *Nanoscale* **2017**, *9*, 965.

[14] E. Tekin, P. J. Smith, U. S. Schubert, *Soft Matter* **2008**, *4*, 703.

[15] H. M. Haverinen, R. A. Myllylä, G. E. Jabbour, *Appl. Phys. Lett.* **2009**, *94*, 073108.

[16] Y. Zou, M. Ban, W. Cui, Q. Huang, C. Wu, J. Liu, H. Wu, T. Song, B. Sun, *Adv. Funct. Mater.* **2017**, *27*.

[17] B. Derby, *Annu. Rev. Mater. Res.* **2010**, *40*, 395.

[18] L. Xie, X. Xiong, Q. Chang, X. Chen, C. Wei, X. Li, M. Zhang, W. Su, Z. Cui, *Small* **2019**, *15*, 1.

[19] H. Shen, Q. Lin, W. Cao, C. Yang, N. T. Shewmon, H. Wang, J. Niu, L. S. Li, J. Xue, *Nanoscale* **2017**, *9*, 13583.

[20] Y. Gong, T. Andelman, G. F. Neumark, S. O'Brien, I. L. Kuskovsky, *Nanoscale Res. Lett.* **2007**, *2*, 297.

[21] M. Shakutsui, H. Matsuura, K. Fujita, *Org. Electron.* **2009**, *10*, 834.

[22] C. Jiang, Z. Zhong, B. Liu, Z. He, J. Zou, L. Wang, J. Wang, J. Peng, Y. Cao, *ACS Appl. Mater. Interfaces* **2016**, *8*, 26162.

[23] J. Sun, B. Bao, M. He, H. Zhou, Y. Song, *ACS Appl. Mater. Interfaces* **2015**, *7*, 28086.

[24] Y. Liu, F. Li, Z. Xu, C. Zheng, T. Guo, X. Xie, L. Qian, D. Fu, X. Yan, *ACS Appl. Mater. Interfaces* **2017**, *9*, 25506.

[25] Y. Park, Y. Park, J. Lee, C. Lee, *J. Appl. Phys.* **2019**,

125, 0.

[26] C. Lin, C. Hung, P. Kuo, M. Cheng, *J. Disp. Technol.* **2010**, *8*, 681.

[27] J. Han, S. I. D. Student, M. Park, Y. Lee, W. K. Bae, **2016**, 545.

[28] X. Xiong, C. Wei, L. Xie, M. Chen, P. Tang, W. Shen, Z. Deng, X. Li, Y. Duan, W. Su, H. Zeng, Z. Cui, *Org. Electron.* **2019**, *73*, 247.

Table. 3 Summary of all the inkjet printed red QLED reported in the literature.

Emitting Materials	Device Structure	HTL	Luminance (cd/m^2)	Current Eff. (cd/A)	Publication date /Ref
CdSe/CdS/ZnS	Normal	poly-TPD	381	N.A	2009/[15]
CdSe/ZnS	Normal	poly-TPD	352	N.A	2010/[26]
CdSe/CdS/ZnCdS	Inverted	CBP	2500	0.29	2016/[27]
CdSe/ZnS	Normal	poly-TPD	12100	4.44	2017/[22]
CdSe/CdS/ZnS	Inverted	PVK	4680	2.54	2017/[22]
CdZnSe/ZnS	Normal	PVK	26320	28.8	2019/[28]
CdZnSe/ZnS	Normal	TFB	2000	0.4	2019/[28]
CdSe/ZnS	Normal	TFB	42000	6.42	2019/This work